VISHNAVI ABHISHIKTA BOLLAPALLI
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EDUCATION

Georgia State University | Atlanta, GA **August 2023 - Present**

- GPA 3.57
- Freshman STEM Student Scholar Award
- Dean's List (Fall 2023, Spring 2024), President's List (Summer 2025, Spring 2026)
- Spring 2024 Chemistry Student of the Semester
- *Relevant Coursework*: Machine Learning, Efficient AI, Application of AI, Software Development, Computer Architecture

SKILLS

Java, Python, C, R | Statistical & ML Modeling: lavaan, jmv (MANCOVA), Latent Growth Curve Modeling, Combat Harmonization | Data Tools: Power BI, Data Visualization, Survey Design, Microsoft Visio | Linux, IT Operations, Microsoft Office Suite

CERTIFICATIONS

Anthropic: AI Fluency Framework & Foundations, Introduction to Model Context Protocol & Advanced Topics, Claude with Google Cloud's Vertex AI | *Codepath*: AI201 Applications of AI

RESEARCH EXPERIENCE

Trends Center- Genetic Risk for Mental Illness and its Relationship to Patterns of Correlated White and Grey Matter
Research Assistant **May 2025 – Present**

- Built R (lavaan) pipeline to fit Latent Growth Curve Models across 69 brain regions from the ABCD study dataset, using MLR/FIML estimation to investigate genetic risk for mental illness and adolescent brain development
- Ran MANCOVA analyses in R (jmv) on 69 brain regions as dependent variables, with sex as a factor and polygenic risk scores plus the Area Deprivation Index as covariates, to test correlations between white and grey matter
- Identified potential biomarkers linked to psychiatric vulnerability through interdisciplinary research
- Contributed to a deeper understanding of how genetic predispositions may influence adolescent brain development.
- Presented research findings to the **University System of Georgia Board of Regents**, showcasing undergraduate research at the state level.

Eye-Tracking Gaze Visualization Research **May 2026 – Present**
Research Analyst

- Built a full gaze visualization pipeline in R processing iMotions eye-tracking exports (fixation/saccade events and 2,042 raw gaze samples at ~120 Hz) across multiple participants and poster stimuli.
- Generated per-participant static outputs (scanpath plots, raw gaze path plots, 2D density heatmaps) and animated GIF gaze replays using ggplot2 and gganimate, with Y-axis coordinate correction for screen-space alignment.
- Engineered automated output organization across 48 participant directories, parsing poster timing from Native_SlideEvents.csv to slice and label gaze data by stimulus.
- Analyzed fixation sequences, saccade amplitudes, and dwell-time distributions to identify visual attention patterns across graphic design poster stimuli.

PROJECTS

FitFindr | AI-Powered Personal Styling Agent
Python, Flask, Groq API, LLMs, Computer Vision, Three.js

- Built an AI fashion recommendation agent that combines wardrobe awareness, thrift item search, outfit generation, and caption creation using LLM-driven planning workflows.
- Integrated Groq-hosted Llama models for natural language styling recommendations and image-based outfit analysis.
- Developed relevance-ranking algorithms and personalized recommendation logic to match clothing items with user preferences and existing wardrobe pieces.
- Created a Flask web application with interactive visualization features and validated functionality through 12 automated Pytest test cases.

FitFin – AI-Powered Personal Finance Simulation Platform

- Built FitFin, a proactive personal finance platform focused on future liquidity forecasting and goal-based budgeting.
- Developed financial simulations to track how purchases affect savings goals and timelines.
- Integrated Amazon Textract OCR for receipt parsing and "Delay-to-Goal" spending analysis.
- Implemented 1,000+ Monte Carlo cash-flow simulations using NumPy/SciPy to predict overdraft and liquidity risk.

- Created explainable AI recommendations with SHAP and InterpretML for transparent financial insights.
- Built scalable backend services with Python, FastAPI, AWS Lambda, and DynamoDB.
- Developed responsive frontend using React, Next.js, TypeScript, Tailwind CSS, and Recharts.

Dead Man's Switch – Secure Automated Trigger System
- Designed and built a high-reliability "Dead Man's Switch" platform that securely releases encrypted data if a user fails to check in within a configurable interval.
- Implemented persistent timer engine using Redis + BullMQ to ensure triggers survive server restarts and prevent false activations.
- Developed zero-knowledge encrypted vault using client-side encryption (Web Crypto API/AES-256) with secure storage in PostgreSQL and AWS S3 (presigned uploads).
- Built backend services with Node.js (Express) and REST APIs, integrating email/SMS notifications (SendGrid/Twilio) with multi-tier reminder and grace-period logic.
- Engineered failure-state protections including audit logs, 2FA authentication, retry queues, health checks, and trigger validation testing to ensure high availability and security.

Custom Processor Design – Computer Architecture
Python, Computer Architecture, Digital Logic, Data Systems
- Designed and implemented a custom processor in Python, defining the instruction set architecture (ISA), data path, and control logic across multiple pipeline stages.
- Built and simulated data system components including registers, ALU, memory modules, and a control unit capable of executing arithmetic, logical, and branching instructions.
- Applied low-level programming and digital logic principles to bridge hardware and software abstraction layers, deepening understanding of processor execution cycles and memory hierarchy.

LEADERSHIP EXPERIENCE

Student Government Organization **Apr 2025 – Apr 2026**
University-Wide Finance Director
- Oversaw financial planning and budget allocation for university-wide student government initiatives across multiple campuses
- Collaborated with executive leadership to ensure fiscal responsibility and transparency in funding decisions
- Streamlined budget proposal processes and supported student organizations in securing funding for impactful programming
- Presented financial reports and recommendations to university stakeholders, advocating for equitable resource distribution
- Led strategic initiatives to optimize funding usage and enhance student engagement through financially sustainable projects

Student Government Association **Mar 2024 – Apr 2025**
Executive Vice President
- Represented the student body in meetings with university faculty and staff, advocating for student interests and concerns.
- Facilitated weekly meetings with club leaders to address their concerns, provide guidance, and ensure effective communication between student organizations and the SGA.
- Oversaw the budgeting and allocation of funds for student organizations, ensuring fair distribution and effective utilization of resources.
- Developed and implemented a comprehensive leadership training program for club executives, enhancing their skills in communication, teamwork, and problem-solving.

CAMPUS EXPERIENCE

IT Support Specialist – Georgia State University Robinson College of Business **May 2026- Present**

- Provided Tier 1 technical support for faculty and students, resolving hardware, software, and network issues in a professional academic setting.
- Guided users on campus technology, including wireless networking and projection systems, to improve overall digital accessibility.
- Configured classroom technology and recording software for lectures and high-profile events, ensuring seamless technical execution.
- Managed Help Desk operations by triaging inquiries, overseeing equipment rentals, and escalating complex issues to senior teams.